

Supplementary Parameters and Comparisons

Transition Based Closed Loop Restoration for Mobility Coupled Distribution Systems with Executable Actions

Scope. Table S1 reports the implemented value, unit, role, calibration rule, selection data, and tested range for manuscript coefficients and execution settings. Long decimal estimates are rounded to eight significant digits for display. Objective weights convert their respective service quantities to a dimensionless score rather than monetary cost.

Table S1: Complete parameters and numerical settings.

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
forecast	L	12	h	observed history length	fixed training configuration	unchanged in strict chronological rerun	not varied
forecast	H	6	h	rollout and forecast horizon	fixed training configuration	unchanged in strict chronological rerun	not varied
forecast	hidden	32	units	graph recurrent hidden width	fixed architecture	unchanged in strict chronological rerun	not varied
forecast	epochs	40	epochs	maximum training epochs	fixed training limit	validation selects the checkpoint within this limit	not varied
forecast	batch	256	samples	training batch size	fixed optimization setting	unchanged in strict chronological rerun	not varied
forecast	learning_rate	0.002	none	AdamW learning rate	fixed training configuration	unchanged in strict chronological rerun	not varied
forecast	weight_decay	0.0001	none	AdamW decoupled weight decay	fixed training configuration	unchanged in strict chronological rerun	not varied
forecast	dropout	0.1	fraction	dropout probability	fixed regularization setting	unchanged in strict chronological rerun	not varied
forecast	epsilon_LN	1e-5	variance unit	LayerNorm denominator stabilizer	torch LayerNorm configuration	all three trained models	not varied
forecast	Huber_delta	1.0	normalized unit	Huber loss transition point	loss evaluated in normalized logarithmic coordinates	2018 to 2021 training targets only	not varied
forecast	gradient_clip	2.0	Euclidean norm	gradient clipping threshold	fixed training configuration	all training updates	not varied
forecast	training_runs	3	integer	independent model fits	fixed repetitions	selected by 2022 validation MAE arrivals plus 0.25 times energy MAE	never selected by test error
forecast	graph_neighbours	4	stations	symmetric geographical nearest neighbour graph	Gaussian distance weighting with self edges and row normalization	station coordinates only	not an observed origin destination graph
objective	gamma	1.0	none	finite horizon discount factor	undiscounted calculation	all scenarios	not varied
objective	lambda_m	1.8	per trip equivalent	mobility loss contribution	declared benchmark weight	fixed before paired evaluation	3.6 in objective sensitivity
objective	lambda_e	2.6	per kWh	unserved energy contribution	declared benchmark weight	fixed before paired evaluation	5.2 in objective sensitivity
objective	lambda_c	1.7	per work-load unit	communication loss contribution	declared benchmark weight	fixed before paired evaluation	3.4 in objective sensitivity
objective	lambda_p	1.2	per kWh	power service contribution	declared benchmark weight	fixed before paired evaluation	2.4 in objective sensitivity
objective	lambda_z	12.0	dimensionless	cross domain exposure contribution	declared benchmark weight	fixed before paired evaluation	6.0 in objective sensitivity
service	beta_m	0.13	relative load	mobility contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied

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Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
service	beta_e	0.19	relative load	energy contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_r	6.0	relative load	road threat contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_p	3.5	relative load	power threat contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	alpha_c	0.40	none	direct communication threat derating	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_mc	0.45	none	communication shortfall to mobility delay	declared benchmark assumption	fixed before paired evaluation	not varied
service	rho_q	0.22	none	backlog carryover	declared benchmark assumption	fixed before paired evaluation	requires less than 1
allocator	kappa_q	1.0	relative weight	backlog in charging score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_ch	1.5	relative weight	current threat in charging score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_E	0.7	relative weight	energy in communication score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_com	1.5	relative weight	current threat in communication score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_p	1.6	relative weight	power threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_c	1.3	relative weight	communication threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_r	1.1	relative weight	road threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_Gamma	0.8	relative weight	multi step marginal benefit	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	eta_service	0.04	fraction	uniform reserve for charging and communication	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	eta_repair	0.02	fraction	uniform reserve for repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	epsilon	1e-9	score unit	square root stabilizer	numerical calculation	fixed	positive by construction
budget	B_ch	18.48321	kWh per h	total hourly charging allocation	1.22 times summed station mean hourly energy	2018 to 2021 Boulder sessions only	stress uses factor 24.4 with 20 times demand
budget	B_com	73.144081	service unit	total communication allocation	0.72 times summed station capacity proxy	proxy uses 2018 to 2021 arrivals and connected session hours	not varied
budget	B_res	10.0	priority unit	total requested restoration priority	max of 10 and 0.018 times summed mean arrivals	2018 to 2021 Boulder sessions only	not varied
transition	rho_r	0.72842984	none	road state persistence	nonnegative deficit regression	2018 to 2021 Boulder station hours	0.54632238 to 0.91053730
transition	sigma	0.23278108	none	spatial propagation	nonnegative deficit regression	2018 to 2021 Boulder station graph	0.11639054 to 0.34917162
transition	rho_p	0.77180562	none	power state persistence	day block bootstrap	aligned EAGLE-I Boulder observations	0.75369266 to 0.78666740
transition	tau_pr	0.0	none	power to mobility association	nonnegative observational regression	aligned EAGLE-I and Boulder EV deficits	0.0 to 0.03180810

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Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
transition	tau_pc	0.42695231	none	power to communication coupling	finite buffer packet construction with hour block bootstrap	12000 scenarios using training period station activity	0.16883429 to 0.55137751
transition	rho_c	0.5	none	communication state persistence	sensitivity only envelope	no longitudinal field packet trace	0.3 to 0.8
transition	tau_rc	0.04	none	road to communication coupling	sign constrained envelope	no joint field trace	0.0 to 0.08
transition	tau_cr	0.08	none	communication to road coupling	sign constrained envelope	no joint field trace	0.0 to 0.16
transition	tau_cp	0.04	none	communication to power service coupling	sign constrained envelope	no joint field trace	0.0 to 0.08
repair	road_effect	0.08	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
repair	power_effect	0.12	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
repair	communication_effect	0.10	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
crew	vehicles	4	integer	available repair crews	declared benchmark setting	all integrated scenarios	12 and 24 in sensitivity
crew	jobs	36	integer	maximum requested work orders per scenario	declared benchmark setting	all integrated scenarios	actual job count depends on the initial threat
crew	speed	30	km per h	road travel speed	declared benchmark setting	Boulder station coordinates	not varied
crew	circuitry	1.25	ratio	road to great circle distance multiplier	declared benchmark setting	Boulder station coordinates	not varied
crew	service_time	empirical	h	job service duration	EAGLE-I 50-customer post peak half recovery prior	255 observed times among 1467 events ending before 2022 clipped to 0.25 to 8 h common route portfolio	county recovery prior rather than measured crew work duration
crew	solver	deterministic list	none	priority per incremental completion time with stable tie break	fixed route rule		no wall clock cutoff
packet	duration	120	s	queue simulation window	declared stress setting	20 training period hours for calibration and main execution response	2023 packet stress reported separately
packet	arrival_rate	8+0.8 activity	packet per s	station traffic arrival intensity	declared stress function	observed connected sessions and arrivals	traffic multipliers 0.5 1 2 and 4
packet	service_rate	30	packet per s	nominal queue service rate	declared stress setting	packet stress simulator	power and communication threat derate it
packet	buffer	100	packets	finite FCFS buffer	declared stress setting	packet stress simulator	not varied
packet	control_fraction	0.05	fraction	share of arrivals carrying control	declared stress setting	packet stress simulator	not varied
packet	deadline	500	ms	on time control deadline	declared stress setting	packet stress simulator	not varied
packet	max_retries	2	integer	retransmission limit	declared stress setting	packet stress simulator	not varied
packet	backoff_mean	0.05	s	exponential retry backoff mean	declared stress setting	packet stress simulator	not varied
packet	erasure	min(0.25,0.002+0.08 z_c)	probability	threat dependent packet erasure	declared stress function	packet stress simulator	not varied
packet	backup	60	s	primary backup duration	declared stress setting	packet stress simulator	0 and 300 s comparisons

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Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
grid	feeder	SMART-DS SFO P1U	case	independent AC execution model	public SMART-DS v1.0 complete feeder	selected before policy comparison	one feeder and 20 mappings
grid	power_factor	0.98	lagging	EV reactive power assumption	declared execution setting	all online OpenDSS actions	not varied
grid	load_model	constant PQ model 1	OpenDSS model	balanced three phase wye EV increment	released eligible load connection and model	17 three phase loads	not varied
grid	voltage_band	0.95 to 1.05	p.u.	accepted active node voltage range	declared operational gate	all online OpenDSS actions	not varied
grid	line_limit	1.0	p.u.	accepted maximum branch loading	declared operational gate	all online OpenDSS actions	not varied
grid	bisection_ iterations	14	integer	common fraction AC projection tolerance	calculation setting	all infeasible raw actions	fraction resolution about 6.1e-5
grid	execution_ convergence_ tolerance	1e-4	OpenDSS criterion	original execution solver convergence	unchanged frozen OpenDSS default	all primary and sensitivity execution	independent measurements checked separately
grid	measurement_ convergence_ tolerance	1e-10	OpenDSS criterion	independent electrical loss and voltage measurement	separate contexts applied to identical accepted loads	all 4096 central and matched pairs	comparison tolerance 1e-8
grid	measurement_ max_ iterations	100	integer	independent measurement iteration limit	precision replay setting	all independent primary measurements	not an algorithm parameter
grid	mappings	20	integer	predefined station to load assignments	geographic rank and fixed initialization	Boulder stations to 17 eligible SMART-DS loads	scenario id modulo 20
grid	mapping_ index	20260825	integer	station mapping initialization index	fixed before evaluation	all mapping assignments	not varied
station_ flow	capacity	max(q0.995*1.2, 7.2)	kWh per h	station service capacity stress proxy	training period empirical quantile	2018 to 2021 Boulder energy	not a charger nameplate claim
station_ flow	closure_ fraction	0.1 0.2 0.3	fraction	closed station stress	declared validation grid	24 selected 2023 hours	full factorial
station_ flow	demand_ multiplier	1 2 4	ratio	demand stress	declared validation grid	24 selected 2023 hours	full factorial
station_ flow	distance_ beta	0.5 1 2	per km	routing distance penalty	declared validation grid	24 selected 2023 hours	full factorial
station_ flow	loyalty	2.0	cost unit	penalty for changing station	declared validation setting	24 selected 2023 hours	not varied
statistics	n_primary	4096	paired scenarios	primary scenario count	prespecified four equal groups	fixed replicate index 2026	1024 per threat class
statistics	bootstrap	20000	resamples	paired mean difference interval	prespecified inference	primary paired scenarios	percentile 95 percent interval
statistics	holm_family	15	tests	total cost primary family	three declared comparators and five groups	overall plus four threat classes	paired t and Wilcoxon adjusted separately
statistics	physical_family	9	tests	secondary service and physical comparisons	separate multiplicity family	central PC versus forecast matched	paired t and Wilcoxon adjusted separately
packet	power_lookup_scale	0.75	normalized availability	distance scale for packet response lookup	declared nearest response rule	training period packet simulations	not varied
packet	communication_lookup_scale	0.375	normalized resource	distance scale for packet response lookup	declared nearest response rule	training period packet simulations	not varied

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Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
packet	communi- cation_rate_ derating	0.70	none	communication threat re- duces the protected service rate	base rate multiplied by 1 minus 0.70 times communi- cation threat	training period packet con- struction	not varied
packet	power_rate_ derating	0.75	none	power threat further re- duces service after backup expires	protected rate multiplied by 1 minus 0.75 times power threat	training period packet con- struction	not varied
packet	minimum_ service_rate	0.5	packets per s	positive queue service floor	applied to protected and degraded rates	all packet simulations	not varied
packet	context_ratio	0.5	none	communication threat relative to power threat in packet construction	constructed context rather than field measurement	training period packet simu- lations	not varied
service	communi- cation_loss_ clip	0.90	fraction	maximum communication shortfall fraction in charg- ing support	clip of communication loss divided by one plus load	continuous score and realized service	not varied
repair	normalized_ priority_cap	3.0	none	maximum predictive repair quantum	requested allocation divided by per zone reference then clipped	continuous priority predic- tion only	not varied
forecast	exposure_ weights	0.25 0.25	relative weights	forecast arrival and energy contributions to exposure	training mean normalization	shared forecast and training means	not varied
forecast	exposure_ bounds	1 to 6	none	exposure clipping range	declared priority scoring range	all priority forecasts	not varied
forecast	training_ mean_floor	1e-6	arrival or kWh	positive floor for exposure normalization	applied to fixed station training means	2018 to 2021 observations only	not varied
crew	job_spread_ weights	0.60 0.30	relative weights	one and two geographical neighbour contributions to initial job priority	common initial threat sum plus weighted spatial expan- sions	shared work requests before policy selection	not varied
crew	job_threshold	1e-10	normalized state	positive initial job inclu- sion threshold	applied before selecting at most 36 jobs	shared work requests before policy selection	not varied
crew	scaled_ duration_ bounds	0.25 to 16	h	service duration bounds after applying the sensitiv- ity multiplier	base empirical prior is first clipped to 0.25 to 8 hours	shared duration draws across policies	service multipliers 0.5 1 and 2
scenario	initial_zone_ counts	2 6 8 12	zones	affected locations in nomi- nal single domain cascade and compound classes	fixed scenario definition	1024 scenarios per class in primary test	shared across policies
scenario	initial_ severities	0.08 0.35 0.45 0.62	normalized state	initial severity in the four classes	fixed scenario definition	nominal single domain cas- cade compound order	shared across policies
scenario	innovation_ scales	0.003 0.008 0.012 0.022	normalized state per h	future disturbance scale in the four classes	fixed scenario definition	nominal single domain cas- cade compound order	unavailable to deploy- able policies
scenario	initial_ multiplier	0.75 to 1.25	ratio	uniform multiplier of ini- tial severity	independent uniform draws on affected locations	shared across policies	not varied
scenario	innovation_ multiplier	0.6 to 1.4	ratio	uniform multiplier of dis- turbance scale	nominal disturbs only the first selected location and other classes all selected locations	shared across policies	not varied
scenario	location_ uniform_mix	0.05	none	positive addition to de- mand weighted location sampling	training mean demand plus 0.05 times its spatial mean then normalized	sampling locations without replacement	not varied
scenario	initial_ threat_cap	0.95	normalized state	initial threat clipping limit	fixed scenario construction	all initial states	not varied

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Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
scenario	transition_ threat_cap	0.98	normalized state	propagated threat clipping limit	fixed transition domain	all prediction and execution transitions	also the nonlinear satu- ration scale
statistics	scenario_ initialization	2026	integer	base initialization index	shared paired random input construction	all frozen primary and sensi- tivity scenarios	not retuned
statistics	scenario_ stride	7919	integer	scenario index multiplier in initialization	base index plus scenario identifier times stride	all frozen primary and sensi- tivity scenarios	not retuned
statistics	coefficient_ noise_index	31991	integer	controller coefficient per- turbation initialization	fixed noisy matrix construc- tion	coefficient noise sensitivity	not retuned

Table S2: Central PC against both service and exposure score controls. Holm correction includes all fifteen comparisons separately for each test. Bootstrap intervals are marginal intervals rather than simultaneous intervals.

Comparator	Group	Pairs	Δ	95% CI	Holm t	Holm Wilcoxon
Forecast matched	All	4096	-1.1436	[-1.5257, -0.7704]	1.38×10^{-8}	9.79×10^{-23}
Forecast matched	Nominal	1024	0.0013	[-0.0042, 0.0075]	1.0000	3.77×10^{-17}
Forecast matched	Single domain	1024	-0.1021	[-0.2602, 0.0510]	0.5860	0.0020
Forecast matched	Cascade	1024	-0.7946	[-1.7624, 0.1327]	0.3944	0.0983
Forecast matched	OOD compound	1024	-3.6789	[-4.8554, -2.5890]	1.94×10^{-9}	1.15×10^{-11}
Exposure matched	All	4096	-7.6977	[-8.3843, -7.0077]	2.83×10^{-97}	5.2×10^{-147}
Exposure matched	Nominal	1024	0.0019	[-0.0037, 0.0081]	1.0000	7.68×10^{-16}
Exposure matched	Single domain	1024	-0.2304	[-0.4169, -0.0533]	0.0626	3.11×10^{-6}
Exposure matched	Cascade	1024	-8.3878	[-9.6902, -7.1245]	2.38×10^{-34}	1.39×10^{-44}
Exposure matched	OOD compound	1024	-22.1746	[-24.3945, -20.0079]	6.17×10^{-72}	1.96×10^{-76}
Exposure central	All	4096	-12.0931	[-13.1616, -11.0704]	1.98×10^{-104}	1.32×10^{-139}
Exposure central	Nominal	1024	0.0103	[0.0047, 0.0165]	0.0036	0.4691
Exposure central	Single domain	1024	-0.3792	[-0.5915, -0.1773]	0.0031	0.1441
Exposure central	Cascade	1024	-8.5773	[-9.9791, -7.2295]	4.84×10^{-32}	2.28×10^{-40}
Exposure central	OOD compound	1024	-39.4260	[-42.9641, -36.0140]	5.55×10^{-89}	1.29×10^{-95}

Statistical comparisons. Table S2 gives the complete primary comparison family. Table S3 reports integrated sensitivities, Table S4 reports physical outcomes, and Table S5 reports empirical cost tails. These tables use the same scenario records and statistical calculations as the main text. Intervals describe simulated scenario uncertainty conditional on the public data and model, not variation across operating utilities. **Execution note.** Continuous restoration values are requested priorities used for candidate scoring. They do not reduce the realized threat state. Packet delivery determines dispatched work orders. The deterministic integer route records completion events. Only those events clear the corresponding realized threat state.

Table S3: All integrated sensitivity comparisons. C denotes central PC, F forecast matched, and R robust PC. The mean difference is first policy minus second policy. Holm correction includes the full sensitivity family. Intervals are marginal.

Condition	Comparison	Pairs	Δ	95% CI	Holm t	Holm Wilcoxon
Primary	R minus C	4096	0.3670	[0.0788, 0.6627]	0.2589	0.1847
20× EV load	C minus F	1024	-3.7920	[-9.9833, 2.2046]	1.0000	1.0000
20× EV load	R minus C	1024	-0.6276	[-6.5897, 5.3242]	1.0000	1.0000
No backup	C minus F	1024	-1.7641	[-2.1646, -1.3886]	2.04×10^{-16}	3.66×10^{-35}
No backup	R minus C	1024	0.1626	[-0.0448, 0.4040]	1.0000	1.0000
60 s backup	C minus F	1024	-1.0259	[-1.6009, -0.4900]	0.0086	7.19×10^{-12}
60 s backup	R minus C	1024	0.2204	[-0.1668, 0.6233]	1.0000	1.0000
300 s backup	C minus F	1024	-1.9467	[-2.8292, -1.0894]	0.0005	1.36×10^{-10}
300 s backup	R minus C	1024	0.8154	[0.0944, 1.5804]	0.5204	0.1050
0.75M	C minus F	1024	3.7779	[2.4720, 5.0919]	8.54×10^{-7}	0.0002
0.75M	R minus C	1024	-3.6909	[-4.7773, -2.6813]	4.01×10^{-10}	9.13×10^{-14}
1.25M	C minus F	1024	-1.2737	[-2.0959, -0.5635]	0.0324	7.69×10^{-7}
1.25M	R minus C	1024	3.4569	[2.4826, 4.5032]	1.02×10^{-9}	4.68×10^{-11}
Coefficient noise 0.15	C minus F	1024	-1.8341	[-2.6349, -1.1321]	6.76×10^{-5}	2.13×10^{-14}
Coefficient noise 0.15	R minus C	1024	1.2651	[0.6919, 1.8547]	0.0006	1.52×10^{-5}
12 crews	C minus F	1024	-0.3849	[-0.4717, -0.3068]	8.38×10^{-18}	1.02×10^{-52}
12 crews	R minus C	1024	0.0279	[-0.0273, 0.0874]	1.0000	1.0000
24 crews	C minus F	1024	-0.6119	[-0.7337, -0.5022]	7.53×10^{-23}	1.49×10^{-74}
24 crews	R minus C	1024	-0.0236	[-0.0538, -0.0003]	1.0000	0.0934
$\rho_c = 0$	C minus F	1024	-1.4042	[-2.0789, -0.7983]	0.0006	8.37×10^{-10}
$\rho_c = 0$	R minus C	1024	0.7161	[0.2249, 1.2241]	0.1195	0.0311
$\tau_{pc} = 0$	C minus F	1024	-1.5121	[-2.1857, -0.9005]	0.0002	3.08×10^{-11}
$\tau_{pc} = 0$	R minus C	1024	0.7820	[0.2952, 1.2915]	0.0562	0.0063
$\tau_{rc} = 0$	C minus F	1024	-1.1054	[-1.9120, -0.3671]	0.1144	0.0007
$\tau_{rc} = 0$	R minus C	1024	0.2731	[-0.3154, 0.8676]	1.0000	1.0000
$\tau_{cr} = 0$	C minus F	1024	-1.4150	[-2.1974, -0.7115]	0.0052	1.18×10^{-7}
$\tau_{cr} = 0$	R minus C	1024	0.5521	[-0.0121, 1.1336]	0.9031	0.4140
$\tau_{cp} = 0$	C minus F	1024	-0.9430	[-1.6911, -0.2523]	0.2088	0.0005
$\tau_{cp} = 0$	R minus C	1024	0.3110	[-0.2648, 0.8932]	1.0000	1.0000
All constructed couplings zero	C minus F	1024	-1.1745	[-1.7176, -0.6815]	0.0003	1.02×10^{-9}
All constructed couplings zero	R minus C	1024	0.7285	[0.2995, 1.1792]	0.0326	0.0009
Constructed couplings doubled	C minus F	1024	0.2837	[-0.7188, 1.2761]	1.0000	1.0000
Constructed couplings doubled	R minus C	1024	-2.0608	[-2.9543, -1.2204]	0.0001	2.23×10^{-6}
Nonlinear realized transition	C minus F	1024	-1.5148	[-1.9601, -1.1040]	3.75×10^{-10}	4.53×10^{-19}
Nonlinear realized transition	R minus C	1024	0.7813	[0.4315, 1.1492]	0.0006	4.6×10^{-5}
2× mobility	C minus F	1024	-1.1782	[-2.0284, -0.3950]	0.1092	0.0006
2× mobility	R minus C	1024	0.3137	[-0.3283, 0.9612]	1.0000	1.0000
2× unserved energy	C minus F	1024	-1.1458	[-2.0653, -0.3006]	0.2088	0.0006
2× unserved energy	R minus C	1024	0.1823	[-0.4827, 0.8508]	1.0000	1.0000
2× communication	C minus F	1024	-1.3283	[-2.3910, -0.3448]	0.2088	0.0003
2× communication	R minus C	1024	0.2559	[-0.5376, 1.0534]	1.0000	1.0000
2× power service	C minus F	1024	-1.4710	[-2.5631, -0.4754]	0.1198	0.0002
2× power service	R minus C	1024	0.3342	[-0.4771, 1.1526]	1.0000	1.0000
0.5× cascade	C minus F	1024	-0.8553	[-1.5909, -0.1857]	0.2760	0.0011
0.5× cascade	R minus C	1024	0.1604	[-0.3922, 0.7155]	1.0000	1.0000

Table S4: Physical outcomes of central PC (C) and forecast matched (F) on 4,096 paired scenarios. Intervals concern C minus F. Holm correction covers all nine endpoints separately for each test. Completion fraction uses requested jobs, including undelivered requests. Scheduled completion and travel include dispatched jobs finishing beyond the horizon. Decision computation time is excluded because concurrent execution is not a controlled timing comparison. Electrical loss and minimum voltage use independent OpenDSS measurements of identical accepted loads at convergence tolerance 10^{-10} , with 10^{-8} comparison solves and at most 100 iterations. These measurements do not alter requests, accepted actions, routes, or costs.

Endpoint	C mean	F mean	Difference 95% CI	Holm t	Holm Wilcoxon
Unserved energy (kWh)	294.358909	294.395760	$[-0.057777, -0.015918]$	0.0037	1.26×10^{-12}
Electrical loss (MWh)	0.841670840	0.841669314	$[6.013 \times 10^{-7}, 2.457 \times 10^{-6}]$	0.0071	2×10^{-14}
Minimum voltage (p.u.)	0.989098473	0.989098647	$[-4.103 \times 10^{-7}, 5.085 \times 10^{-8}]$	0.7135	0.0117
Timely action fraction	0.976412	0.976401	$[-2.077 \times 10^{-5}, 4.310 \times 10^{-5}]$	1.0000	1.0000
Retained action fraction	1.000000	1.000000	$[0, 0]$	1.0000	1.0000
Curtailment (kWh)	0	0	$[0, 0]$	1.0000	1.0000
Completed/requested jobs	0.811998	0.810978	$[5.779 \times 10^{-4}, 0.001470]$	6.51×10^{-5}	0.0013
Scheduled completion (h)	3.096674	3.118857	$[-0.026427, -0.018020]$	2.11×10^{-23}	8.59×10^{-18}
Scheduled travel (h)	2.621301	2.619267	$[-0.003585, 0.007646]$	1.0000	1.0000

Table S5: Primary empirical cost distribution. Quantiles and upper tail means describe the same 4,096 scenarios per policy. Tail means include a fractional observation at their probability boundary. These descriptive values are not inferential evidence of tail superiority.

Policy	Mean	95% quantile	99% quantile	Upper 5% mean	Upper 1% mean	Maximum
Forecast matched	1113.68	2599.56	3334.41	3062.58	3694.88	4494.26
Central PC	1112.54	2597.87	3331.67	3055.50	3687.60	4494.26
Robust PC	1112.90	2603.73	3331.44	3058.35	3692.33	4494.26
Exposure matched	1120.23	2618.63	3357.45	3082.80	3729.44	4531.67
Exposure central	1124.63	2632.39	3369.16	3105.46	3746.92	4507.25
Static	1202.74	2767.64	3507.76	3224.38	3895.21	4799.52
Greedy	1122.54	2632.59	3372.35	3093.79	3744.89	4494.26
Oracle	1053.67	2582.47	3301.28	3019.15	3647.82	4474.21